Multi-Area Frequency Security Region with Dynamic Tie-Line Mutual Assistance

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***Abstract*—As power systems move toward low-inertia operation, frequency security increasingly depends on where inertia and damping resources are located. Existing single-area H-D support sets identify feasible inertia-damping bounds, but they do not capture dynamic assistance through inter-area tie-lines. This letter proposes a multi-area frequency security region (MA-FSR) framework for quantifying regional operating bounds under coupled frequency dynamics. The framework combines a decoupled analytical approximation with a state-space dynamic search that accounts for tie-line capacity and synchronizing stiffness limits. Case studies on the IEEE two-area Kundur system show that dynamic mutual assistance can substantially expand feasible H-D regions, especially in the disturbed area. The MA-FSR therefore provides a compact way to evaluate how interconnection strength changes regional frequency-security margins.**

***Index Terms*—System frequency response, multi-area power systems, tie-line mutual assistance, frequency security region.**

I. Introduction

THE rapid integration of converter-interfaced generation (CIG) is reducing the rotational inertia available in modern power grids. Frequency-security constraints, including rate of change of frequency (RoCoF) and frequency nadir, have therefore become binding limits in unit commitment, dispatch, and planning studies [1]–[4]. The single-area H-D support set [5] provides a compact representation of feasible equivalent inertia (H) and damping (D) parameters. However, practical power systems are organized as interconnected control areas. Their regional frequency responses are reshaped by tie-line power exchange, which cannot be represented by an isolated single-area model.

This letter extends the frequency security region concept to multi-area power systems. During a contingency, the disturbed area can import power from healthy areas, thereby reducing its local frequency excursion. We study this effect using two complementary evaluation methods. Option A is a decoupled analytical approximation that treats tie-line exchange as an equivalent reduction in step disturbance. Option B is a dynamic simulation-based bisection search that preserves the coupled multi-state interaction and the nonlinear tie-line capacity limit. The resulting multi-area frequency security region (MA-FSR) is visualized in the H-D plane, and the effects of tie-line capacity (C_{12}) and synchronizing stiffness (T_{12}) are quantified. The simulation repository is open-sourced at https://github.com/Divas1234/frequencyregions.git.

II. Multi-Area Frequency Dynamics & Modeling

Consider a two-area interconnected power system in which Area 1 is disturbed by a step power imbalance ΔP_1, while Area 2 provides dynamic mutual assistance through the tie-line.

A. Option A: Decoupled Approximation

In the decoupled approach, tie-line exchange is approximated as a static contribution to the area power imbalance. For Area i connected to neighboring areas, the effective disturbance $\Delta P_{eff,i}$ is defined as

$$\Delta P_{eff,i} = \Delta P_i + \sum_{j\in\Omega_i} \kappa_{ij} C_{ij} \tag{1}$$

where C_{ij} is the tie-line transmission capacity, and $\kappa_{ij} \in [-1, 1]$ represents the direction and magnitude of the assisting power flow. The single-area analytical bounds in [5] are then evaluated with $\Delta P_{eff,i}$ to construct decoupled H-D regions.

B. Option B: Dynamic State-Space Modeling

To capture the transient interaction, the interconnected system is represented by a five-state model. For each area $i \in \{1, 2\}$, the coupled swing equation is

$$2H_i\dot{\Delta f_i} = \Delta P_{m,i} - D_i\Delta f_i - \Delta P_{dist,i} + \Delta P_{tie,i} \tag{2}$$

where $\Delta P_{tie,1} = \Delta P_{tie}$ and $\Delta P_{tie,2} = -\Delta P_{tie}$. The governor-turbine systems are modeled as:

$$\Delta P_{m,i} = -K_{m,i}\Delta f_i + x_{g,i} \tag{3a}$$

$$\dot{x}_{g,i} = \frac{1}{T_{g,i}}\left[-(1/R_i - K_{m,i})\Delta f_i - x_{g,i}\right] \tag{3b}$$

where $T_{g,i}$ is the turbine-governor time constant, R_i is the droop gain, and $K_{m,i}$ is the high-pressure fraction. The tie-line power flow is governed by

$$\dot{\Delta P}_{tie} = 2\pi f_0 T_{12}(\Delta f_2 - \Delta f_1) \tag{4}$$

subject to the capacity saturation constraint $|\Delta P_{tie}| \le C_{12}$, where T_{12} is the synchronizing coefficient and $f_0 = 50$ Hz is the nominal frequency.

Defining the state vector $x = [\Delta f_1, x_{g,1}, \Delta f_2, x_{g,2}, \Delta P_{tie}]^T$ and input vector $u = [\Delta P_{dist,1}, \Delta P_{dist,2}]^T$, the unsaturated dynamics can be written in state-space form as

$$\dot{x} = Ax + Bu \tag{5}$$

where the state matrix A and input matrix B are

$$A = \begin{bmatrix} -\frac{D_1+K_{m,1}}{2H_1} & \frac{1}{2H_1} & 0 & 0 & \frac{1}{2H_1} \\ -\frac{1/R_1-K_{m,1}}{T_{g,1}} & -\frac{1}{T_{g,1}} & 0 & 0 & 0 \\ 0 & 0 & -\frac{D_2+K_{m,2}}{2H_2} & \frac{1}{2H_2} & -\frac{1}{2H_2} \\ 0 & 0 & -\frac{1/R_2-K_{m,2}}{T_{g,2}} & -\frac{1}{T_{g,2}} & 0 \\ -2\pi f_0 T_{12} & 0 & 2\pi f_0 T_{12} & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} -\frac{1}{2H_1} & 0 \\ 0 & 0 \\ 0 & -\frac{1}{2H_2} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}^T$$

When the tie-line power flow reaches the transmission limit, ΔP_{tie} is clamped to $\pm C_{12}$ and the corresponding state derivative $\dot{\Delta P}_{tie}$ is set to zero.

III. Multi-Area Frequency Security Region

The MA-FSR is defined as the set of regional inertia (H_i) and damping (D_i) parameters that satisfy both local and system-wide frequency-security constraints:

$$\max|\Delta f_i(t)| \le \gamma_{Nadir,i}, \quad \forall i \tag{6a}$$

$$\max|\Delta \dot{f}_i(t)| \le \gamma_{RoCoF,i}, \quad \forall i \tag{6b}$$

$$\zeta_i \le 1.0, \quad \forall i \tag{6c}$$

Because the initial RoCoF occurs at $t = 0^+$, before the tie-line flow deviates, the local RoCoF boundary remains independent of tie-line parameters:

$$H_{i,RoCoF} = \frac{\Delta P_{dist,i}}{2\gamma_{RoCoF,i}} \tag{7}$$

Following the single-area H-D support-set derivation in [5], the dynamic feasibility of Area i can be separated into local stability bounds and frequency-performance bounds. Let D_i be fixed during boundary tracing. The damping-dependent stability interval is written as

$$H^-_{i,stab}(D_i) = \frac{T_{g,i}}{2}\left(\eta_i - \sqrt{\eta_i^2 - (D_i + K_{m,i})^2}\right), \tag{8a}$$

$$H^+_{i,a}(D_i) = \frac{T_{g,i}}{2}\left(\eta_i + \sqrt{\eta_i^2 - (D_i + K_{m,i})^2}\right), \tag{8b}$$

$$H^+_{i,b}(D_i) = \frac{T_{g,i}}{2}(D_i + K_{m,i}), \tag{8c}$$

$$H^+_{i,stab}(D_i) = \min\Big[H^+_{i,a}(D_i),\, H^+_{i,b}(D_i)\Big], \tag{8d}$$

where $\eta_i = D_i - K_{m,i} + 2/R_i$. These bounds are inherited from the damping-ratio condition $0 \le \zeta_i \le 1$ and prevent the search from accepting frequency trajectories that satisfy nadir limits but violate local modal damping requirements.

The nadir boundary is denoted by $H_{i,Nadir}(D_i)$. Under Option A, it is obtained by evaluating the closed-form single-area nadir expression with $\Delta P_{eff,i}$. Under Option B, it is obtained from the coupled time-domain trajectory. The lower and upper MA-FSR envelopes are therefore

$$H^-_i(D_i) = \max\Big[H_{i,RoCoF},\, H^-_{i,stab}(D_i),\, H_{i,Nadir}(D_i)\Big], \tag{9a}$$

$$H^+_i(D_i) = H^+_{i,stab}(D_i). \tag{9b}$$

The area is feasible at damping D_i only if $H^-_i(D_i) \le H^+_i(D_i)$. This construction turns each regional frequency-security requirement into an explicit geometric boundary in the H-D plane.

To compute the MA-FSR boundary under Option B, a bisection search is performed:

1) For a given damping value D_i in Area i, initialize the regional inertia interval $[H_{min}, H_{max}]$.
2) At $H_{mid} = \frac{1}{2}(H_{min} + H_{max})$, simulate the five-state model using fourth-order Runge-Kutta integration.
3) Track $\Delta f_j(t)$ for all areas $j \in \{1, 2\}$ and test the nadir condition $\max|\Delta f_j(t)| \le \gamma_{Nadir,j}$.
4) If both areas satisfy the nadir constraints, set $H_{max} = H_{mid}$; otherwise, set $H_{min} = H_{mid}$.
5) The search terminates when $|H_{max} - H_{min}| < \epsilon$ (where $\epsilon = 10^{-3}$), yielding the critical inertia limit $H_{i,crit}(D_i)$.

A quadratic regression is then fitted:

$$H_{i,crit}(D_i) = c_0 + c_1 D_i + c_2 D_i^2 \tag{10}$$

The fitted $H_{i,crit}(D_i)$ is used as $H_{i,Nadir}(D_i)$ in (9). The feasible operating region is then obtained by sweeping D_i and retaining the interval between $H^-_i(D_i)$ and $H^+_i(D_i)$.

IV. Case Studies and Analysis

The IEEE two-area Kundur test system is used for verification. Both areas have nominal droop gains $1/R_i = 33.33$ p.u. and turbine parameters $T_{g,i} = 0.25$ s and $K_{m,i} = 0.35$. The nadir and RoCoF thresholds are $\gamma_{Nadir,i} = 0.5$ Hz and $\gamma_{RoCoF,i} = 0.5$ Hz/s. Area 1 is subjected to a local step contingency of $\Delta P_1 = 3.5$ p.u., while Area 2 has $\Delta P_2 = 2.0$ p.u. The nominal tie-line parameters are $T_{12} = 4.0$ p.u. and $C_{12} = 1.0$ p.u.

A. MA-FSR Expansion under Interconnection

Fig. 1 compares the feasible operating regions of Area 1 and Area 2 in the H-D plane. The isolated case uses $C_{12} = 0$, whereas the interconnected case uses $C_{12} = 1.0$ p.u. In Area 1, which experiences the heavier disturbance, interconnection shifts the nadir boundary downward. At nominal damping, the required inertia decreases from approximately 99.6 s to 59.5 s, which substantially enlarges the feasible region. In Area 2, the interconnected nadir boundary reaches the search floor of 0.05 s because Area 2 is supported by Area 1's nominal inertia ($H_1 = 8.0$ s). Its feasible region is therefore governed mainly by the local RoCoF requirement, $H_2 \ge 4.0$ s.

B. Impact of Tie-Line Capacity and Stiffness

Fig. 2 shows how Area 1's security region changes with tie-line capacity C_{12} and synchronizing stiffness T_{12}. As C_{12} increases from 0 to 4.0 p.u. (Fig. 2a), the critical inertia boundary decreases progressively. Once C_{12} exceeds the peak tie-line power flow of approximately 2.5 p.u., additional capacity no longer expands the feasible region. Increasing T_{12} (Fig. 2b) accelerates inter-area power transfer, reduces the transient frequency drop in Area 1, and further enlarges the feasible H-D region.

V. Conclusion

This letter extends the frequency security region framework to multi-area power systems. By combining a decoupled approximation with a nonlinear dynamic simulator, the proposed MA-FSR characterizes regional H-D operating bounds under tie-line mutual assistance. The IEEE two-area Kundur case shows that interconnection can substantially relax the disturbed area's inertia requirement, while the benefit saturates when tie-line capacity exceeds the peak assisting power flow. The MA-FSR provides a compact tool for coordinating regional frequency-control parameters and assessing multi-area scheduling security under explicit tie-line limits.

References

[1] S. A. Hosseini, M. Toulabi, A. S. Dobakhshari, A. Ashouri-Zadeh, and A. M. Ranjbar, "Delay compensation of demand response and adaptive disturbance rejection applied to power system frequency control," *IEEE Transactions on Power Systems*, vol. 35, no. 3, pp. 2037–2046, 2020.

[2] G. Tian and Q. Z. Sun, "A stochastic controller for primary frequency regulation using on/off demand side resources," *IEEE Transactions on Smart Grid*, vol. 14, no. 5, pp. 4141–4144, 2023.

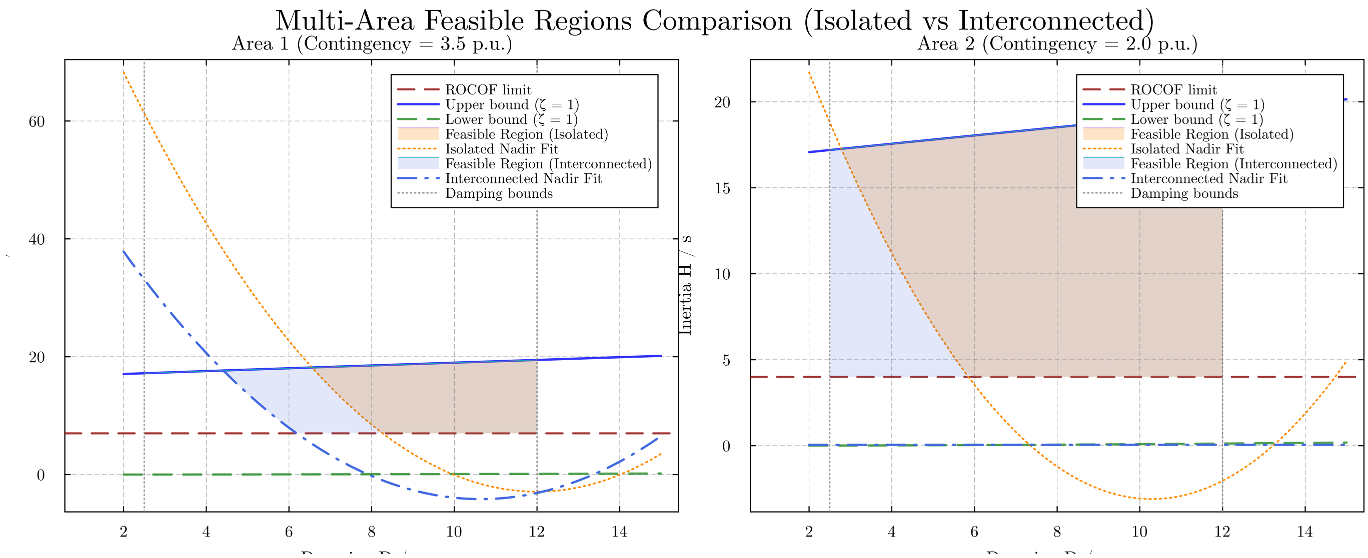
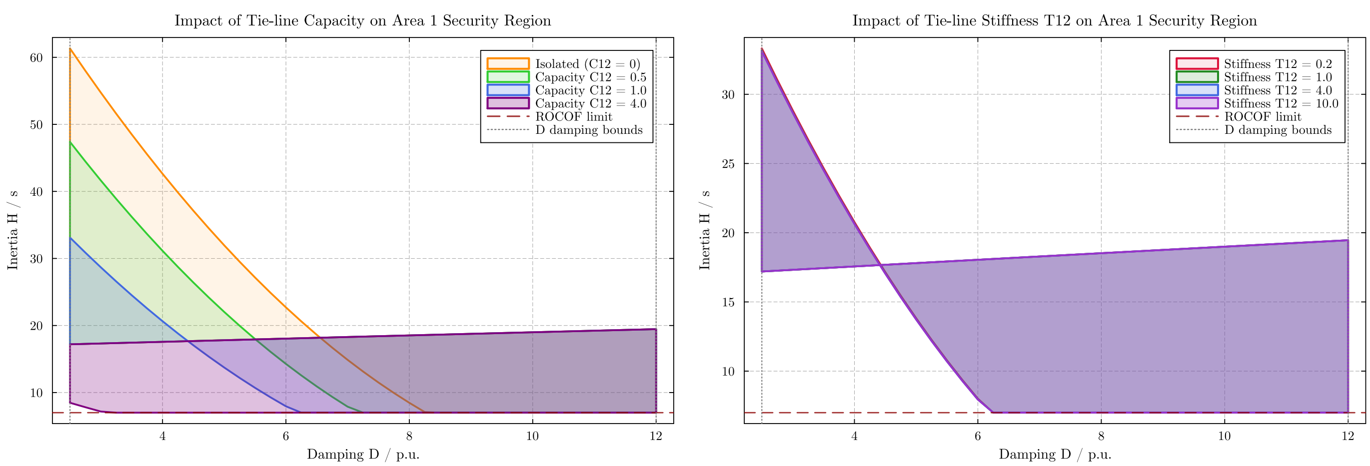


Fig. 1. MA-FSR comparison between isolated and interconnected states under Option B.



(a) Tie-line capacity sweep (b) Stiffness sweep

Fig. 2. Impact of tie-line capacity C_{12} and stiffness T_{12} on Area 1's feasible boundary.

[3] U. Markovic, V. Häberle, D. Shchetinin, G. Hug, D. Callaway, and E. Vrettos, "Optimal sizing and tuning of storage capacity for fast frequency control in low-inertia systems," in *2019 International Conference on Smart Energy Systems and Technologies (SEST)*. IEEE, 2019, pp. 1–6.

[4] Y. Yuan, Y. Zhang, J. Wang, Z. Liu, and Z. Chen, "Enhanced frequency-constrained unit commitment considering variable-droop frequency control from converter-based generator," *IEEE Transactions on Power Systems*, vol. 38, no. 2, pp. 1094–1110, 2023.

[5] Y. Yuan, L. Zhu, Z. Zhang, W. Hu, and Z. Chen, "h-d support set: Identifying frequency dynamics by constraining multidimensional frequency parameter bounds," *IEEE Transactions on Power Systems (under review)*, 2026.